

Calculus I Practice Final — Worked Answer Key

A balanced twenty-five-question cumulative final with point values, study map, and complete solutions.

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1. Evaluate $\lim_{x \rightarrow 2}(x^2 + 3x - 1)$.

Answer: 9

- a. Direct substitution applies.
- b. The verified result is 9.

2. Evaluate $\lim_{x \rightarrow 4}(x^2 - 16)/(x - 4)$.

Answer: 8

- a. Factor and cancel $x-4$.
- b. The verified result is 8.

3. Choose k so $f(x) = kx + 1$ for $x < 2$ and $f(x) = 7$ for $x \geq 2$ is continuous.

Answer: $k = 3$

- a. Match the left limit $2k+1$ to 7.
- b. The verified result is $k = 3$.

4. Use the limit definition to find the derivative of x squared.

Answer: $2x$

- a. Expand $(x+h)$ squared, cancel, divide by h , and let $h \rightarrow 0$.
- b. The verified result is $2x$.

5. Differentiate $4x^5 - 3x^2 + 7$.

Answer: $20x^4 - 6x$

- a. Apply linearity and the power rule.
- b. The verified result is $20x^4 - 6x$.

6. Differentiate $x^2 \sin x$.

Answer: $2x \sin x + x^2 \cos x$

- a. Use the product rule.
- b. The verified result is $2x \sin x + x^2 \cos x$.

7. Differentiate $(x + 1)/(x - 1)$.

Answer: $-2/(x - 1)^2$

- a. Use the quotient rule and simplify.
- b. The verified result is $-2/(x - 1)^2$.

8. Differentiate $(3x^2 + 1)^4$.

Answer: $24x(3x^2 + 1)^3$

a. Outer power times inner derivative.

b. The verified result is $24x(3x^2 + 1)^3$.

9. For $x^2 + y^2 = 25$, find dy/dx .

Answer: $-x/y$

a. Differentiate both sides with respect to x .

b. The verified result is $-x/y$.

10. Find the tangent line to $y = x^3$ at $x = 2$.

Answer: $y - 8 = 12(x - 2)$

a. Evaluate the point and derivative.

b. The verified result is $y - 8 = 12(x - 2)$.

11. Find c guaranteed by MVT for $f(x) = x^2$ on $[1, 3]$.

Answer: $c = 2$

a. Set $2c$ equal to the secant slope 4.

b. The verified result is $c = 2$.

12. Find critical numbers of $x^3 - 3x$.

Answer: $x = \pm 1$

a. Solve $3x^2 - 3 = 0$.

b. The verified result is $x = \pm 1$.

13. Where is $x^3 - 3x$ increasing?

Answer: $(-\infty, -1) \cup (1, \infty)$

a. Use the sign of $3(x^2 - 1)$.

b. The verified result is $(-\infty, -1) \cup (1, \infty)$.

14. Find the inflection point of $x^3 - 6x^2$.

Answer: $(2, -16)$

a. $f'' = 6x - 12$ changes sign at 2.

b. The verified result is $(2, -16)$.

15. A rectangle has perimeter 40. Find maximum area.

Answer: 100

a. $A = x(20 - x)$ is maximal at $x = 10$.

b. The verified result is 100.

16. A circle radius grows at 2 cm/s. Find dA/dt at $r=5$.

Answer: $20\pi \text{ cm}^2/\text{s}$

- Differentiate $A=r$ squared with respect to time.
- The verified result is $20\pi \text{ cm}^2/\text{s}$.

17. Use linearization at 9 to approximate $\sqrt{9.2}$.

Answer: $3 + 0.2/6 \approx 3.0333$

- $L(x)=3+(x-9)/6$.
- The verified result is $3 + 0.2/6 \approx 3.0333$.

18. Find $\int (6x^2 - 4)dx$.

Answer: $2x^3 - 4x + C$

- Integrate term by term.
- The verified result is $2x^3 - 4x + C$.

19. Evaluate $\int_0^2 3x^2 dx$.

Answer: 8

- Use x cubed at the endpoints.
- The verified result is 8.

20. Differentiate $F(x) = \int_1^x \cos(t^2)dt$.

Answer: $\cos(x^2)$

- Apply FTC Part I.
- The verified result is $\cos(x^2)$.

21. Evaluate $\int 2x(x^2 + 1)^3 dx$.

Answer: $(x^2 + 1)^4/4 + C$

- Let $u=x$ squared+1.
- The verified result is $(x^2 + 1)^4/4 + C$.

22. Find the area under $y=x$ on $[0,3]$.

Answer: $9/2$

- Integrate x or use triangle area.
- The verified result is $9/2$.

23. Find the average value of x squared on $[0,3]$.

Answer: 3

- Divide the integral 9 by interval length 3.
- The verified result is 3.

24. If $v(t)=3t^2-6t$, find displacement from 0 to 3.

Answer: 0

- a. Integrate velocity over the interval.
- b. The verified result is 0.

25. Explain why a differentiable function is continuous.

Answer: *The derivative limit forces $f(x) - f(a)$ to approach 0.*

- a. Factor $f(x)-f(a)$ as a difference quotient times $x-a$.
- b. The verified result is *The derivative limit forces $f(x) - f(a)$ to approach 0..*